US corn yield in August: a scored 80% band

Desk memo. Public USDA data only, rebuilt as it stood on each release morning. Every number below is generated by the repository's `make memo`.

Question

How far can USDA's January final corn yield land from its August print? We put an 80% band around the August print, built from USDA's own past August-to-January misses, and scored it on 22 seasons, 2004–2025. It covered 21 of 22; the pre-registered verdict is "too wide".

August 2025 answer

Object	Unit	USDA point	q10	q50	q90	80% band	Truth (Jan first-final)	Inside
M1	bu/ac	188.8	178.6	188.8	200.6	[178.6, 200.6]	186.5	yes
PROD	Mbu	16742	15592	16768	17893	[15592, 17893]	17021	yes

On August 12, 2025, the USDA estimated corn yield at 188.8 bu/ac, with our 80% band ranging from 178.6 to 200.6 bu/ac. The corresponding production estimate was 16,742 Mbu, with our band at 15,592 to 17,893 Mbu. By January, the final yield was 186.5 bu/ac and production was 17,021 Mbu, both within their respective bands. This single season's result is merely a check on the arithmetic, with the true test of our band's performance lying in its 22-season track record.

Stocks-to-use decision line

USDA's August balance sheet puts 2025/26 stocks-to-use at 13.3%. Our q10 production, put through that sheet with use held fixed, gives 6.1% (q50 13.4%, q90 20.5%). This is conditional on USDA's use projections and is not scored. Price steepens at low stocks (Wright 2011), so the downside tail is the one that matters.

How the band scored, 2004–2025

Band	Horizon	Level	n	x	Coverage	CP90	Pre-registered verdict
Yield	May	80%	22	20	90.9%	[74.1%, 98.4%]	not distinguishable
Yield	Jun	80%	22	18	81.8%	[63.1%, 93.5%]	not distinguishable
Yield	Jul	80%	22	18	81.8%	[63.1%, 93.5%]	not distinguishable
Yield (primary)	Aug	80%	22	21	95.5%	[80.2%, 99.8%]	too wide
Yield	Sep	80%	22	21	95.5%	[80.2%, 99.8%]	too wide
Yield	Oct	80%	20	17	85.0%	[65.6%, 95.8%]	not distinguishable
Yield	Nov	80%	22	17	77.3%	[58.0%, 90.6%]	not distinguishable
Production	Aug	80%	22	20	90.9%	[74.1%, 98.4%]	not distinguishable

The primary pre-registered result shows that 21 out of 22 seasons fell within the August band, resulting in a 95.5% coverage rate. The 90% Clopper-Pearson interval for this coverage rate is 80.2%–99.8%, with the pre-registered wording describing the band as "too wide". The estimator used is conservative by construction, so at the nominal 80% rate it is expected to cover 84.2% of seasons at these sample sizes; that accounts for about 4 points of the excess coverage. Adding crop condition (Good/Excellent share) made no detectable difference to the August band (one-sided p = 0.888). Out of 33 exploratory tests, none had Holm-adjusted p-values below 0.05.

When this fails

The band breaks in two failure modes: drought springs and late surprises. In 2012, a drought year, the bands from the May, June and July origins missed the final, which fell beyond both the 50% and 80% bands; the August band covered it. In 2020 the January final fell outside the 80% band from the October and November origins. The stocks-to-use line holds use fixed, so in years like 2011 and 2012 its q10 would need negative stocks and gives no number. Eleven WASDE and nine NASS releases survive only as corrected copies, with the originals not archived; they are flagged, and no USDA error is claimed.

Port to other origins

The code is reused as is, with nothing Russia-specific, on Russian wheat production. This measures how far USDA revises its own Russia number before settling it in April, about three years after harvest. It does not measure the harvest. The line shows the harness runs on a second series and makes no claim about effectiveness. Analyzing 18 seasons, from 2006 to 2023, the August USDA print's distance to the settled figure shows an 80% band covering 18 out of 18 seasons, with a 90% Clopper-Pearson interval of 84.7% to 100% coverage. The verdict on the band is "too wide". The estimator alone expects 84.4% coverage at 18 seasons, so about 4 points of the excess coverage come from it. This exploratory analysis can be applied to other origins, such as EU rapeseed or Brazil safrinha, where dated vintages exist (MARS, CONAB).